

Paper Manager pack — ACT / ALOHA (Oct 12)

Paper: Learning Fine-Grained Bimanual Manipulation with Low-Cost Hardware

arXiv: <https://arxiv.org/abs/2304.13705> · RSS 2023

Site: <https://tonyzhaozh.github.io/aloha/>

PDF: /workspace/papers/act-aloha-2304.13705.pdf

Session: Oct 12 · due 9:30 AM · Empiricist=3, Archaeologist=5; others empty — full five packs

Sources: Summarizer + Senior Research (/workspace/act-aloha-quiet/DELIVERABLE.md)

1. Paper summary

ALOHA = <\$20k leader–follower bimanual teleop (WidowX → ViperX, 4 RGB, 50 Hz). **ACT** = ~80M CVAE+Transformer predicting **action chunks** ($k \times 14$ absolute joints) + **temporal ensembling**. Fine contact-rich skills from ~10–20 min demos (e.g. Open Cup 84%, Slot Battery 96%, Ziploc 88%, Shoe 92%).

Club line: Parallel 2023 chunk ancestor with Diffusion Policy → $\pi_0/\pi_{0.5}$ FM chunks → RTC → BEHAVIOR.

2. Keywords

ALOHA, ACT, action chunking, temporal ensembling, CVAE, bimanual teleop, absolute joints @ 50 Hz, Diffusion Policy lineage, π_0 chunks, RTC, BEHAVIOR

Default: $k=100$ (~2s @ 50 Hz); $L1+\beta KL$ ($\beta=10$); test $z=0$; TE every-step query + exp-weighted overlap.

3. Motion / control spine (IMPORTANT — do not conflate)

Scheme	Mechanism
ACT TE	Every-step query; exp-avg overlapping preds for same t

DP receding	Predict T_p, exec T_a<T_p, replan
π₀ open-loop	H=50; exec prefix; **no TE** (hurt early for them)
RTC soft-inpaint	Async; freeze prefix; soft-mask remainder
BEHAVIOR'25	Predict 30 / exec 26 / keep 4 + correlated FM

CLEAR: chunking is shared; TE ≠ receding ≠ open-loop ≠ soft-inpaint.

Map: $k/50 \leftrightarrow \text{DP } \tau_a \cdot \Delta t \leftrightarrow \pi_0/\text{Comet } H/\text{Hz}$.

4. Role packs

Stakeholder (18+3)

Cheap bimanual data engine + chunking control prior. Not a foundation model (per-task ~80M). Follow-ons: Mobile ALOHA, ALOHA 2. 2026 BEHAVIOR stacks inherit chunk horizons/overlap from this fork.

Scientific Reviewer (8+3)

Strengths: hardware+algo synergy; hard tasks with subtask tables; ablations for k/TE/CVAE; honest failures.

Critiques: real 1 seed×25 thin; BeT/RT-1 without chunks in main tables; z=0 discards multimodality; no language; 30 fps cams vs 50 Hz control underspecified.

Verdict: Right cite for action chunking + low-cost bimanual teleop.

Empiricist (11+3) — weight 3

H: Mid chunk (~2s) beats k=1 and episode-long; TE reduces jerk (+~3pp paper); CVAE critical for human demos, ≈noop for scripted.

Priority	Experiment
P0 Y	$k \in \{1,10,50,100,200,400\}$ (expect peak ~100); TE on/off
P0 Partial	CVAE on/off human vs scripted
P1	Chunking on BC-MLP; target vs delta; L1 vs L2
Defer	Full real ALOHA 6-task / paper seed protocol

Budget: ~8–12 GB; quiet P0–P1 ≤40–60 GPU-h; sim Transfer Cube first. Env: /workspace/act-aloha-quiet/env_outline/.

Tattoo: $k=1 \rightarrow 100$ success **1% → 44%**; CVAE on human **35.3% → 2%** without.

Archaeologist (11+3) — HIGH weight 5

Priors: BC/ALVINN; DAgger compounding; Lai chunking psych; Transformers/BERT; CVAE/ β -VAE; BeT/RT-1/VINN; joint-space teleop; ResNet.

Pivot: popularized action chunking + TE for high-Hz visuomotor BC on open low-cost bimanual HW.

Subsequent: DP || chunks+receding; Mobile ALOHA; ALOHA 2; π_0 cites ACT (FM chunks, no TE); $\pi_{0.5}$; RTC; BEHAVIOR'25/'26.

Genealogy: ALOHA+ACT (CVAE+TE) || DP (diffusion+receding) → Mobile/ALOHA2 → $\pi_0/\pi_{0.5}$ → RTC → BEHAVIOR'25 forks → BEHAVIOR'26.

Visionary (11+3)

Chunking non-negotiable; CVAE+TE optional. Prefer RTC/receding over naïve full open-loop. Start OpenPI $\pi_{0.5}$ / GR00T N1.7; port horizon/overlap/correlated noise — not CVAE. **Durable idea:** short generative action trajectories as motor primitive; generative family churns; **chunk + overlap execution** stays. Skip Discord/Drive.

5. Slide Maker brief

Role	Timing	Focus
Stakeholder	18+3	Cheap teleop + chunking prior
Reviewer	8+3	Cite hygiene + thin real n
Empiricist	**11+3**	**k-sweep + TE + CVAE (HIGH)**
Archaeologist	**11+3**	**Chunk genealogy to BEHAVIOR (HIGH)**
Visionary	11+3	Chunk+overlap endures

Spine: action chunking as motor contract; TE vs receding vs open-loop table. PDF for Mac.

GitHub title: [ECE 605] - ACT Making Club Pack

Paths

Artifact	Path
This pack	`/workspace/act-aloha-quiet/KANTA_PACK.md`
Senior Research	`/workspace/act-aloha-quiet/DELIVERABLE.md`

Summarizer	`/workspace/papers/act-aloha-2304.13705-summary.md`
Paper PDF	`/workspace/papers/act-aloha-2304.13705.pdf`